.:• Frontier

Brineworks

Carbon dioxide removal prepurchase application

2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](https://github.com) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Brineworks B.V.

Company or organization location (we welcome applicants from anywhere in the world)

Amsterdam, Netherlands

Name(s) of primary point(s) of contact for this application

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Brief company or organization description <20 words

Brineworks is pioneering next-generation electrochemical CDR with our low-cost electrolyzer that provides pure CO_2 from air without additional post-processing

1. Public summary of proposed project[1] to Frontier

a. **Description of the CDR approach:** Describe how the proposed technology removes CO_2 from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your

[1] We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

Overview

Brineworks is developing a new class of electrochemical carbon removal technology engineered from the ground up for permanent, affordable, and scalable CO_2 drawdown. At the core of our system is a patented hybrid electrolyzer that uniquely combines the high-throughput benefits of water electrolysis with the ionic selectivity of electrodialysis. Our ground-breaking innovation allows us to produce acid and base at high rates, using virtually any low-cost salt as a working electrolyte in a closed-loop pH-swing process. Atmospheric CO_2 is captured and regenerated as a high-purity gas >99% without additional need for purification, which can be readily injected via geological sequestration for high-permanence removal.

Our system is modular, fully electrified, and operates entirely without heat or engineered sorbents. We've already demonstrated that we can cycle our system on and off for years at a time without accelerating degradation, which means our process is directly compatible with intermittent renewable energy. It also means that our modular architecture will unlock flexible siting across industrial and commercial locations, from near-coast to far-inland operating environments. Critically, every ton of CO_2 captured is compatible with underground mineral storage, which ensures that Brineworks will deliver high-durability carbon removal at globally relevant cost and scale.

How the System Works

The Brineworks process removes atmospheric CO_2 by leveraging the bicarbonate equilibrium in aqueous solutions. When exposed to air, alkaline solutions absorb CO_2 and convert it to dissolved bicarbonate and carbonate, which is a process governed by well-established carbonate chemistries (shown below) and Henry's Law. This contacting solution becomes more bicarbonate-rich over time, and is then fed through our hybrid electrolyzer which electrochemically splits saltwater streams into acidic and basic streams. The acidic chamber of our device shifts the local pH experienced by bicarbonate ions to a lower value, which chemically converts them into dissolved CO_2 gas which can be easily extracted from the water. That gas is then stripped under mild conditions using a commercial separation vessel. Finally, it is collected, dried, and routed to permanent storage.

$$CO_{2(atmosphere)} \leftrightarrow CO_{2(aq)}$$

$$CO_{2(aq)} + H_2O_{(l)} \leftrightarrow H_2CO_{3(aq)} \leftrightarrow H^+_{(aq)} + HCO^-_{3(aq)} \leftrightarrow 2H^+_{(aq)} + CO^{2-}_{3(aq)}$$

Following gas removal, the process is restarted. The alkaline stream generated by the electrolyzer, now with a very high capacity for CO_2 absorption, is delivered toward a commercial air contactor to replace the CO_2 that was previously stripped using the acidic stream. Because all electrochemistry takes place in a closed ionic loop, and because no solid sorbents or chemical additives are consumed, the working fluids remain stable across thousands of cycles. The CO_2 is isolated with >99% purity, making it directly compatible with downstream compression, transport, and injection. The entire process is electric, membrane-light, and can be operated at both ambient temperature and pressure without the need for any solids handling.

What Makes Brineworks Different

Brineworks departs from both traditional and electrochemical Direct Air Capture (DAC) systems in several important ways. Unlike solid-sorbent DAC, our system is not limited by the extreme energy penalty that comes with thermal desorption, nor by the high cost associated with expensive and unstable amine materials. In addition, our stack can achieve significantly higher current densities for acid and base production, at **far lower capex**, and with more efficient use of electricity than conventional electrodialysis-based electrochemical DAC systems. Because we avoid reliance on high-cost fluorinated membranes, precious metals (Ir, Ru, Pt), and other expensive metals (e.g. Ti) the entire platform is built for manufacturability and low-cost multiplication.

Another key differentiator is our operational flexibility, which actually stems from our ability to thrift out precious metals that are unstable over long periods of intermittency, in favor of low-cost Earth-abundant materials that are thermodynamically stable at both the “on” and “off” operating points in our system. In other words, our electrolyzer is designed to handle full intermittency with no loss in performance. In testing, we have already demonstrated multiple years’ worth of start-stop cycles without accelerating material degradation. This flexibility allows us to run when electricity is clean and cheap, avoiding both the need for grid firming and the risk of fossil backstopping. It is immediately obvious that the ability to truly run on behind-the-meter power will come from our ultra-low CapEx designs where off-time does not destroy the payback economics of an installation, and where load cycling does not introduce additional avenues for material degradation. The Brineworks technology satisfies both of these requirements.

Our system has the optionality to co-produce green hydrogen as a byproduct of the water-splitting step, although this is not essential to our business model, and for CDR-specific installations can be entirely toggled off to save on energy cost and simplify off-take logistics. Where hydrogen is locally useful—such as in synthetic fuel hubs or industrial clusters—it may be separately valorized. Irrespective of this green hydrogen co-production flexibility, our DAC system is designed to yield cost-competitive CO_2 for sequestration.

Most importantly, Brineworks is focused on building the carbon removal technology of the future. We are agnostic to alternative end-uses of our highly scalable technology platform, which already include CO_2 sequestration, carbon avoidance, utilization, and offsetting. We aim to reduce the amount of CO_2 in the atmosphere by a gigaton and beyond. That is our ultimate mission.

Scalability and Deployability

Our technology is built to scale safely and quickly, without the constraints that have historically limited other engineered CDR pathways. Our system does not require biomass, mineral feedstocks, or other consumable chemical inputs, so we are not limited by land-use, crop cultivation rates, or supply chains. The modular nature of our system allows us to deploy across existing industrial corridors and coastal energy hubs (which will facilitate the transport of liquified CO_2 for sequestration, if transport is indeed necessary at large deployment scales), with zero environmental disruption and no use of arable land.

Our platform is also uniquely suited to take advantage of high-Dissolved Inorganic Carbon (DIC) aqueous equilibria. Whereas traditional amine sorbent-based DAC operates with 420 ppm CO_2 in air which is in natural competition with H_2O in the air for binding sites over the sorbent, our pH-swing approach dramatically improves thermodynamic favorability of CO_2 extraction from the air. This translates to an energy intensity of as low as ~1.2 MWh per ton of CO_2 captured, enables higher capture rates from any given volume of air processed, and allows for deployment across a range of geographies with fluctuating temperature and humidity conditions without degrading the system or complicating plant control systems.

Cost Outlook and Pathway to <$100/ton

The Brineworks system was designed to address the primary cost drivers of electrochemical CDR: CapEx and energy consumption. Our hybrid electrolyzer achieves significantly higher acid/base output per m^2 of membrane used than conventional electrodialysis, which reduces capital intensity by an estimated **~$32/t$CO_2$**. By maximizing membrane utilization and entirely avoiding expensive materials, we further reduce cost at scale compared to other high-current electrolyzer geometries because our electrolyzer is not reliant on ultra-high-cost precious metals and fluorinated membranes. In comparison to state-of-the-art high-current electrolyzers in the CDR space, our installed electrolyzer CapEx is 10% lower, with Brineworks replacement part costs levelized by tonnage of CO_2 captured over the lifetime of a plant coming in at a mere 48% of the competition.

Our energy use is highly competitive with almost all other CDR pathways, ranging from 1.2 to 1.8 MWh per ton of CO_2 removed (including air contacting and CO_2 compression energy) depending on the configuration and deployment climate. At projected electricity prices of $50/MWh which is already regularly achievable for green electricity around the world, our system projects to deliver removal at **$89/t$CO_2$**, even before any forward-thinking learning rates are applied. With learning effects and volume manufacturing, we anticipate reaching sub-$100/ton all-in cost by the end of this decade—and well below that at GtCO_2/y scale as wind/solar prices continue to decrease. In optimistic renewable energy deployment scenarios, we are projecting a cost range of **$35-75/t$CO_2$**, even when doubling or tripling installed capacity of equipment to maintain nameplate output with low up-time when running on wind and solar. Our cost-down trajectory is a testament to our engineering philosophy which is to leverage ultra-low CapEx, intermittency-stable electrolyzers to ride the wind and solar cost curves into the future.

Alignment with Frontier's Priority Innovation Areas

Brineworks directly addresses several of Frontier's 2025 target criteria. Our approach is fully electrochemical, does not require any thermal energy, and has already demonstrated that it is compatible with intermittent renewables. It delivers near-term, verifiable CO_2 removal with a system that is already field-tested and technologically on track to support first deliveries in 2026. Our projected cost curve is steep and yet highly conservative; it is underpinned by predictable engineering and supply chain creation rather than speculative materials science advances or external interventions.

Our system is also infrastructure-adaptable. It can deploy modularly alongside existing facilities to reduce marginal costs and environmental risks from those facilities, or it can be deployed independently up to MtCO_2/y scale. Because our system generates high-purity CO_2 gas in a closed-loop process, the removal is also incredibly easy to track, meter, and validate through third-party validated MRV.

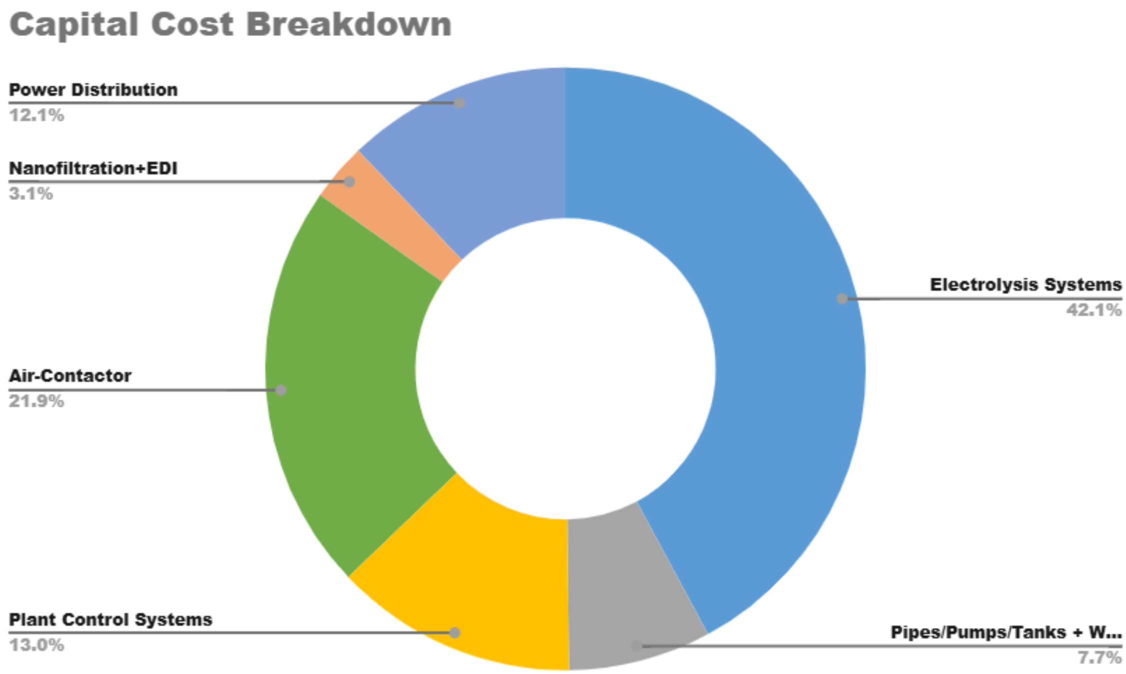
b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?[2] What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

Our current project details: For the proposed project, we are installing an initial 25 tCO_2/y capacity DAC module in Amsterdam where we have already secured a site at the Matrix Innovation Center in the Amsterdam Science Park. We will operate that 25 tCO_2/y module for the first year of the project (2026) with CDR deliveries to Frontier. In Q2 2026, we will expand on the 25 tCO_2/y module to a total DAC capacity of 175 tCO_2/y (i.e. a 150 tCO_2/y capacity increase with Frontier pre-purchase support). In parallel with the increase in module capacity, the electrochemical stack capacity will be increased to meet the entire 175 tCO_2/y capacity of the project. The project will operate at the 175 tCO_2/y capacity for two additional years (Q2 2026-Q2 2028), giving a total gross CDR delivery of 375 tons over a running period of 30 months, with an effective price per ton to Frontier of $2,193/t. We expect to continue operation of this site as a test-bed for process improvements indefinitely.

In this project, we will compress and transport the CO_2 to one of three potential 44.01 injection sites, and will work in collaboration with 44.01, ClimatePoint and Isometric to verify the credits. We are currently discussing contract terms with Isometric, and have established an engagement with both ClimatePoint for carbon accounting verification as well as 44.01 for geologic sequestration, as 44.01 can fully comply with Isometric requirements for verification (as well as others e.g. Puro.earth). We have already prepared the engineering documentation for our system and have an ongoing relationship with the Dutch engineering firm ChemXpro who will provide the front-end engineering and design package for the scale-up modules, as well as operate as our EPCM partner.

Our cost breakdown: Full engineering documentation for the plant has been prepared in collaboration with ChemXpro based in The Netherlands. We have started the process of equipment procurement, and are now in the early stages of finalizing purchase prices for the capacity scale-up. A Class III cost estimate has already been performed, with the ratiometric cost breakdown given in the accompanying pie chart based on our conservative estimate of final capital costs. Current cost-per-ton projections are dominated by the small scale at which we are manufacturing the electrolysis system, although these will come down significantly in the long term as supply chains are secured, economies of scale are established, and stack manufacturing is at least partially verticalized.

[2] We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at $100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.



How we get to scale: In order for our core electrolysis technology to scale, in the short term (<2030) we must:

1) Improve our cost-advantage over competitive electrochemical systems by leveraging our competitive energy spend and favorable system CapEx, to further drive down levelized CDR cost.
2) Encourage faster industrial scale-up of currently off-the-shelf but under-produced electrolyzer components like ion-exchange membranes.
3) Secure a stack manufacturing supply chain to minimize scale-up risk by verticalizing on production wherever is reasonable.

Our approach to quantifying CO_2 removed: Considering the nature of our process which will involve direct capture of atmospheric CO_2, we directly measure the mass of CO_2 removed from air (CDR_M), and collaborate with sequestration operators to monitor injection site leakage (Sequestration Efficiency) so that we fully understand exactly how much CO_2 is transported between air and Earth. This monitoring is done with calibrated mass-flow meters and purity sensors, with a secondary measurement coming from direct gravimetric analysis of compressed CO_2 prior to injection.

Gross Removal Including Sequestration: $\mathbf{CDR_G}$ = $\mathbf{CDR_M}$ x Sequestration Efficiency (%)

Our gross removal of CO_2 (CDR_G) is the product of what we directly extract from the atmosphere, corrected for sequestration efficiency, which in the case of our injection sequestration partners, is close to 100%. Net removal is then this gross removal value subtracted by our Scope 1-3 emissions, as determined both internally and through a third-party LCA. Net emissions include transportation emissions, end-of-life disposal emissions, and emissions from energy production and transmission.

c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

There are several important risks to consider for our CDR approach. These include:

Technical Risk:
The core electrochemical system at the heart of our DAC process relies on stable, repeated cycling of electrolytes for the acid/base generation involved with pH swing capture. While our stack has already undergone over 1,000 successful start-stop cycles without affecting performance degradation, scaling from prototype-scale to full plant operation carries some inherent technical risks. These include the potential for membrane damage, unanticipated current efficiency losses, or stack imbalances under periodic high-throughput conditions. We are mitigating these risks by using only earth-abundant, electrochemically stable electrode materials that do not require precious metal "dimensionally stable" coatings which are susceptible to thermal/mechanical stress cracking and subsequent electrochemical corrosion during load cycling. We also minimize membrane use relative to traditional electrodialysis technologies, and we integrate real-time cell-by-cell monitoring to identify failure points in the stack more quickly.

Project Execution Risk:
As with any novel infrastructure project, execution risk exists around supply chain delays, permitting bottlenecks, and integration challenges during commissioning. We have partnered with an experienced EU-based engineering firm to guide procurement and early-stage deployment and are executing under a Class III cost estimate with defined procurement workflows. This approach reduces interdependency risks during the commissioning phase.

MRV Risk:
Direct air capture projects must demonstrate that every ton of CO_2 claimed as removed is both physically captured and permanently stored. For looping DAC systems like ours, the primary MRV challenge is in verifying that CO_2 captured into solution has been fully liberated and stored, without re-emission. Our process design includes direct metering of gas flows and mass balances at all critical stages: bicarbonate formation, acidification and gas liberation, compression, and transfer to storage. Purity sensors, flowmeters, and pH control systems will be integrated to ensure transparency and traceability. High built-in confidence in the measurement is physically ensured by the fact that the CO_2 into the contacting solution can only have come from the atmosphere. In addition, we are working with third-party Life Cycle Assessment (LCA) and MRV firms to validate our monitoring plan and ensure conformance with emerging standards for DAC protocols that involve mineralization as the sequestration pathway. We signed an LOI with Climate Point and are already assessing with them options to improve our current LCA and GHG accounting. We have also engaged in discussions with Isometric as our preferred verification provider, with the context being our partnership with 44.01, since both companies are also open to collaborating.

Ecosystem Risk:
Because our DAC process is a closed-loop, electrically powered system that does not interact with seawater, soil, or biota, we carry virtually no direct ecological disturbance risk. Our process fluid remains entirely internal to the system, and all gaseous products are either captured for storage or vented safely (high-purity O_2 gas as the sole potential waste product). Nevertheless, we are proactively engaging environmental experts to evaluate any potential for material leaks, noise, or localized emissions during operation and will meet or exceed all relevant permitting thresholds in any host region, not just the

thresholds in place in The Netherlands.

Financial Risk:
While our DAC economics are structured to be viable without reliance on co-product sales or subsidies, early-stage deployment still carries financial risk. This includes both capital cost uncertainty and operational variability due to fluctuating electricity prices. To mitigate this, we prioritize future siting in regions with stable, low-carbon power grids (e.g., Iceland, southern Europe, MENA) and are exploring fixed-rate energy procurement agreements where possible. Additionally, our modular design allows for staged CapEx commitment and learning-driven iteration, reducing the potential financial burden of scaling prematurely.

Policy and Market Risk:
CDR remains a policy-dependent market. While Frontier and other advanced market commitments provide valuable early demand, long-term project bankability will be influenced by carbon accounting rules, certification pathways, and policy incentives (e.g., 45Q, EU ETS inclusion). We actively track relevant policy developments in the EU, MENA, and North America, and are designing our project documentation and MRV stack to align with the most robust and durable standards. We also anticipate that by producing high-purity CO_2 with independently verified storage, our DAC projects will be well-positioned under any future compliance regime.

d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) *(should be net volume after taking into account the uncertainty discount proposed in 5c)*	248
Delivery window *(at what point should Frontier consider your contract complete? Should match 2f)*	2028
Levelized cost ($/ton CO_2) *(This is the cost per ton for the project tonnage described above, and should match 6d)*	2,193
Levelized price ($/ton CO_2)[3] *(This is the price per ton of your offer to us for the tonnage described above)*	2,193

[3] This does not need to exactly match the cost calculated for "This Project" in the TEA spreadsheet (e.g., it's expected to include a margin and reflect reductions from co-product revenue if applicable).